Complete Tech Engineer Reference

50 Categories | 300+ Commands | Priority Ordered

File System • Networking • Security • Storage • Process Management
Monitoring • Scripting • Kernel • Forensics • Server Administration

PRIORITY LEGEND			
CRITICAL	HIGH	IMPORTANT	STANDARD
Use every day	Use most days	Regular work	Periodic use

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01 File and Directory Management

Foundation of everything — navigate, create, move, delete

CRITICAL

Command	Description	Key Flags / Notes
`ls`	List directory contents	`-l long -a hidden -h human -R recursive -t time`
`cd`	Change Directory — navigate the filesystem	`cd ~ home cd - previous cd .. parent`
`pwd`	Print Working Directory — show current path	
`mkdir`	Make Directory — create new folders	`-p create parents -m set permissions`
`rmdir`	Remove empty directory	
`rm`	Remove files or directories	`-r recursive -f force -i interactive -v verbose`
`cp`	Copy files or directories	`-r recursive -p preserve -u update -v verbose`
`mv`	Move or rename files and directories	`-i interactive -f force -v verbose`
`touch`	Create empty file or update timestamp	`-t timestamp -m modify time -a access time`
`ln`	Create hard or symbolic links	`-s symbolic -f force -v verbose`
`find`	Search files by name size date type perms	`-name -type -size -mtime -exec -perm`
`locate`	Fast file search using index database	`-i case insensitive -n limit results`
`tree`	Visual directory structure in tree format	`-L level -a hidden -d dirs only -h human`
`file`	Detect and display the type of a file	`-b brief -i mime type`
`stat`	Detailed file metadata — size perms timestamps	`-f filesystem -c format`
`realpath`	Resolve and print absolute path of a file	
`basename`	Extract filename from a full path	
`dirname`	Extract directory portion from a full path	
`rename`	Bulk rename files using patterns	`-n dry run -v verbose`
`which`	Show full path of an executable command	`-a show all matches`
`whereis`	Find binary source and man page of command	`-b binary only -m manual only`

Command	Description	Key Flags / Notes
type	Show how the shell interprets a command	-a show all -t type only

02 File Viewing and Reading

CRITICAL

Read logs, configs, scripts — daily necessity

Command	Description	Key Flags / Notes
cat	Concatenate and print file content to terminal	-n number lines -A show special chars -s squeeze blanks
less	Scrollable file viewer with search (q to quit)	/search n next N prev G end g start
more	Basic pager — older alternative to less	-d show help -s squeeze +n start at line n
head	Show first N lines of a file (default 10)	-n number -c bytes -q quiet
tail	Show last N lines — tail -f for live log follow	-n number -f follow -F follow rename -c bytes
tac	Print file in reverse line order	
od	Octal dump — view binary file in octal hex	-x hex -c chars -b octal -A address
hexdump	Hex viewer for binary files	-C canonical -n bytes -s skip
strings	Extract readable text strings from binary files	-n min length -a scan all
wc	Count words lines characters in a file	-l lines -w words -c bytes -m chars
diff	Compare two files line by line	-u unified -i ignore case -r recursive -q brief
cmp	Compare two files byte by byte	-l verbose -s silent
comm	Compare two sorted files line by line	-1 suppress col1 -2 suppress col2 -3 suppress col3

03 Text Processing and Parsing

CRITICAL

Process logs, extract data, transform text — the engineer's core skill

Command	Description	Key Flags / Notes
grep	Search text using patterns (Global Regex Print)	-i case insensitive -r recursive -n line number -v invert -l files -c count -E extended regex
egrep	Extended grep — supports + ? | patterns	-i -r -n same as grep -E
sed	Stream Editor — find replace transform text	-i in place -n suppress -e expression -f file s/old/new/g

Command	Description	Key Flags / Notes
awk	Column-based text processing and reporting	-F field separator -v variable NR line num NF num fields
cut	Extract specific columns or fields from text	-d delimiter -f fields -c characters -b bytes
paste	Merge lines from multiple files side by side	-d delimiter -s serial
join	Join lines from two files on a common field	-t delimiter -1 field1 -2 field2 -i ignore case
sort	Sort lines of text alphabetically or numerically	-n numeric -r reverse -k key -t delimiter -u unique -h human
uniq	Remove or count duplicate adjacent lines	-c count -d only dupes -u only unique -i ignore case
tr	Translate or delete characters in text	-d delete -s squeeze -c complement
tee	Write to file AND stdout simultaneously	-a append -i ignore interrupt
xargs	Build and execute commands from standard input	-n max args -I replace -P parallel -0 null delimiter
column	Format text output into aligned columns	-t table -s separator -c width
expand	Convert tabs to spaces	-t tab size
fmt	Format text to a specified line width	-w width -u uniform spacing
nl	Number lines of a file	-b style -v start num -i increment

04 File Searching and Filtering

Find anything on the system instantly

CRITICAL

Command	Description	Key Flags / Notes
find	Search files by name size date type permissions	-name -type f/d -size -mtime -perm -exec -maxdepth
grep	Search inside file contents using patterns	-r recursive -l filenames -n line num -i case -v invert
locate	Fast file search using updatedb index	-i case insensitive -n limit -r regex
updatedb	Update the locate command file database	
which	Show full path of a command	-a all matches
whereis	Find binary source and man page of command	-b binary -m manual -s source
type	Show how shell interprets a command name	-a all -t type -p path

Command	Description	Key Flags / Notes
xargs	Pipe output as arguments to another command	-I{} replace -n count -P parallel -0 null

05 File Permissions and Ownership

CRITICAL

Control who can read write execute — critical for security

Command	Description	Key Flags / Notes
chmod	Change file permissions (read write execute)	-R recursive +x add exec 755 644 777 u g o a
chown	Change file owner and group	-R recursive user:group -h symlinks
chgrp	Change file group ownership	-R recursive -v verbose
umask	Set default permission mask for new files	022 default 002 group write 077 private
getfacl	Get Access Control List of a file	-R recursive -p skip comments
setfacl	Set Access Control List of a file	-m modify -x remove -R recursive -b remove all u:user:rwx
chattr	Change file attributes (immutable append-only)	+i immutable +a append -R recursive -v verbose
lsattr	List file attributes set by chattr	-R recursive -a hidden -l long

06 Process Management

HIGH

Control what runs on the system — kill hang debug processes

Command	Description	Key Flags / Notes
ps	List running processes (Process Status)	-aux all processes -ef full format -p pid --forest tree
top	Live interactive process monitor	k kill r renice q quit M sort mem P sort cpu 1 per-cpu
htop	Better interactive process monitor (color UI)	F5 tree F9 kill F6 sort F4 filter
kill	Send signal to process by PID	-9 SIGKILL force -15 SIGTERM graceful -1 SIGHUP reload
killall	Kill all processes matching a name	-9 force -i interactive -u user -v verbose
pkill	Kill processes matching a pattern	-f full command -u user -9 force -signal

Command	Description	Key Flags / Notes
pgrep	Find process PID by name or pattern	-l show name -a full cmd -u user -f full cmd
nice	Start process with custom CPU priority	-n value (-20 highest to 19 lowest)
renice	Change priority of already running process	-n value -p pid -u user -g group
nohup	Run process immune to terminal hangup/logout	nohup cmd & output goes to nohup.out
jobs	List background jobs in current shell	-l show PID -r running -s stopped
fg	Bring background job to foreground	fg %1 fg %jobname
bg	Continue stopped job in background	bg %1
wait	Wait for background process to finish	wait PID wait %job
watch	Repeat a command every N seconds	-n interval -d highlight diff -t no title
time	Measure how long a command takes to run	time command real user sys output
timeout	Kill command automatically after N seconds	--signal=KILL -s signal type
strace	Trace system calls made by a process	-p PID -e trace=open -o file -f follow forks
lsof	List open files and which process uses them	-p PID -i port -u user -n no DNS +D dir
fuser	Find which process is using a file or port	-k kill -m mount -n network -v verbose

07 User and Group Management

Multi-user systems — create manage control access for users

HIGH

Command	Description	Key Flags / Notes
whoami	Show current logged-in username	
who	Show all users currently logged into system	-b last boot -H headers -q count
w	Show logged-in users and what they are doing	-h no header -s short -u show idle
id	Show UID GID and group memberships of a user	id username
groups	Show groups the current user belongs to	groups username
useradd	Create a new user account	-m create home -s shell -g group -G extra groups -u uid

Command	Description	Key Flags / Notes
usermod	Modify existing user account settings	-aG add group -s shell -L lock -U unlock -d home
userdel	Delete a user account	-r remove home -f force
groupadd	Create a new group	-g gid -r system group
groupmod	Modify an existing group	-n rename -g new gid
groupdel	Delete a group	
passwd	Change user password	-l lock -u unlock -d delete -e expire -S status
su	Switch User — become another user	su - full login shell su username
sudo	Run a command as superuser or another user	-u user -i login shell -s shell -l list perms
visudo	Safely edit the sudoers file with syntax check	
chsh	Change the login shell for a user	-s shell -l list shells
chfn	Change user full name and contact info	
last	Show history of user logins and reboots	-n count -F full time username
lastlog	Show most recent login of all users	-u user -t days
finger	Show user info — name login shell home dir	finger username

08 System Monitoring and Performance

HIGH

Know CPU RAM disk I/O in real time — essential for server work

Command	Description	Key Flags / Notes
top	Live CPU memory process monitor	1 per-core M mem sort P cpu sort k kill
htop	Color interactive process and resource viewer	F6 sort F4 filter F9 kill F5 tree
vmstat	Virtual Memory Statistics — RAM swap I/O CPU	-s summary -d disk -n header once interval count
iostat	CPU and disk Input/Output statistics	-x extended -d disk only -c cpu only interval
mpstat	Per-CPU statistics for multi-core analysis	-P ALL per cpu -u utilization interval
sar	System Activity Reporter — historical stats	-u cpu -r mem -d disk -n network -q load

Command	Description	Key Flags / Notes
`free`	Show RAM and swap memory usage	`-h human -m MB -g GB -s repeat -t total`
`uptime`	Show system uptime and load averages	`1min 5min 15min load averages`
`dmesg`	Kernel ring buffer messages — boot and hardware	`-T timestamps -l level -w follow --clear`
`lscpu`	CPU architecture and core information	
`lsmem`	Show memory blocks and their online state	
`nproc`	Print number of available CPU processing units	
`lshw`	Detailed hardware information listing	`-short -class network -html -json`
`lspci`	List all PCI devices connected to system	`-v verbose -k show kernel driver -nn ids`
`lsusb`	List all USB devices connected to system	`-v verbose -t tree -s bus:dev`
`dmidecode`	Read hardware info from BIOS/UEFI DMI table	`-t type --type memory --type processor`
`perf`	Linux performance analysis and profiling tool	`stat record report top list`
`sysctl`	Read and modify kernel parameters at runtime	`-a all -w write -p load file net.ipv4.ip_forward`

09 Networking and Connectivity

Configure interfaces check connectivity manage routes

HIGH

Command	Description	Key Flags / Notes
`ip`	Configure network interfaces routes tunnels	`addr link route neigh rule tunnel maddr`
`ifconfig`	Interface configuration — older but still used	`-a all up/down netmask broadcast`
`ping`	Test connectivity to a host using ICMP	`-c count -i interval -s size -t ttl -W timeout`
`ping6`	Test IPv6 connectivity to a host	`-c count -i interval`
`ss`	Socket Statistics — modern replacement of netstat	`-t tcp -u udp -l listening -p process -n numeric -a all`
`netstat`	Network connections routing stats (older)	`-t tcp -u udp -l listen -p program -r routing -n numeric`
`traceroute`	Trace the network path packets take to host	`-n no DNS -m max hops -w wait -I ICMP`

Command	Description	Key Flags / Notes
`mtr`	Combines ping and traceroute in real time	`--report -n no DNS -c count --tcp`
`curl`	Transfer data from/to URLs — HTTP FTP etc	`-o output -O remote name -L follow -I header -X method -d data -H header -u auth`
`wget`	Download files from web via HTTP HTTPS FTP	`-O output -r recursive -q quiet -c continue --limit-rate`
`hostname`	Show or set the system hostname	`-I all IPs -f FQDN -s short`
`hostnamectl`	Query and change system hostname (systemd)	`set-hostname status`
`route`	Show and modify the IP routing table	`-n numeric add del default gw`
`arp`	View and modify ARP cache table	`-a all -d delete -s static -n no DNS`
`ip route`	Modern routing table management	`show add del default via dev`
`ethtool`	Query and control network driver and hardware	`-i driver -s speed -k offload -S stats`
`iwconfig`	Configure wireless network interfaces	`essid key mode freq channel`
`nmcli`	NetworkManager command line interface	`con show dev status con up/down radio wifi`

10 SSH and Remote Access

Securely connect manage and transfer files on remote servers

HIGH

Command	Description	Key Flags / Notes
`ssh`	Secure Shell — encrypted remote login	`-p port -i key -L local forward -R remote forward -D dynamic -N no cmd -v verbose`
`scp`	Secure Copy — transfer files over SSH	`-r recursive -P port -i key -v verbose -C compress`
`sftp`	Secure FTP — interactive file transfer	`-P port -i key get put ls cd mkdir`
`ssh-keygen`	Generate SSH public/private key pairs	`-t rsa/ed25519 -b bits -C comment -f filename -p change passphrase`
`ssh-copy-id`	Copy public key to remote server authorized_keys	`-i key -p port user@host`
`ssh-agent`	Authentication agent — holds private keys	`eval $(ssh-agent) ssh-add`
`ssh-add`	Add private key identity to ssh-agent	`-l list -D delete all -t lifetime`

Command	Description	Key Flags / Notes
rsync	Fast efficient file sync local or remote	-a archive -v verbose -z compress -P progress --delete -n dry run --exclude
telnet	Test raw TCP port connectivity (insecure)	telnet host port
nc	Netcat — raw TCP/UDP connections and testing	-l listen -p port -v verbose -z scan -u UDP -e exec

11 Service and Daemon Management

Start stop restart enable disable services on the server

IMPORTANT

Command	Description	Key Flags / Notes
systemctl	Control systemd services units targets	start stop restart reload enable disable status is-active list-units daemon-reload
journalctl	Query and view systemd service logs	-u unit -f follow -n lines --since --until -p priority -b boot -k kernel
service	SysV init service control (older)	start stop restart status reload
init	Change system runlevel (older SysV)	0 halt 1 single 3 multi 5 gui 6 reboot
runlevel	Show current and previous runlevel	
systemd-analyze	Analyze system boot performance time	blame critical-chain plot verify
loginctl	Manage systemd login sessions users	list-sessions list-users show-session terminate-session
timedatectl	Query and change system time and date	set-time set-timezone set-ntp list-timezones status
localectl	Query and change system locale settings	set-locale set-keymap list-locales status
hostnamectl	Query and change system hostname	set-hostname status set-icon-name

12 Disk and Storage Management

Partition format mount manage drives — server storage ops

IMPORTANT

Command	Description	Key Flags / Notes
df	Disk Free — show space usage per partition	-h human -T filesystem type -i inodes -a all
du	Disk Usage — show size of files and directories	-h human -s summary -a all --max-depth -c total

Command	Description	Key Flags / Notes
`lsblk`	List block devices — drives and partitions	`-f filesystem -o output -t tree -m permissions`
`fdisk`	Partition management for MBR disks	`-l list -u units device path p print n new d delete w write`
`parted`	Partition management for GPT and MBR disks	`-l list mkpart rm resizepart print align-check`
`gdisk`	GPT partition table editor	`-l list device p print n new d del w write`
`mount`	Mount a filesystem to a directory	`-t type -o options -a all from fstab -l list -v verbose`
`umount`	Unmount a mounted filesystem	`-l lazy -f force -v verbose`
`blkid`	Show UUID type label of block devices	`-s UUID -o format device`
`dd`	Low-level block copy and disk imaging	`if=input of=output bs=block size count status=progress conv=sync`
`sync`	Flush filesystem buffers to disk	
`fstab`	File /etc/fstab — configure auto mount at boot	`UUID mountpoint fstype options dump pass`
`hdparm`	Get and set hard disk drive parameters	`-I info -t speed test -s power -S standby -g geometry`
`smartctl`	Check disk health using S.M.A.R.T. data	`-i info -H health -a all -t short/long test`
`badblocks`	Scan for bad sectors on a block device	`-v verbose -w write test -n non destructive`

13 Filesystem Management

Create repair tune filesystems — ext4 xfs btrfs

IMPORTANT

Command	Description	Key Flags / Notes
`mkfs`	Format partition with a filesystem type	`-t ext4/xfs/btrfs/vfat -L label -b block size`
`mkfs.ext4`	Create ext4 filesystem (most common Linux)	`-L label -b block -m reserved -j journal`
`mkfs.xfs`	Create XFS filesystem (high performance)	`-L label -b size -f force`
`mkfs.btrfs`	Create Btrfs filesystem (modern CoW)	`-L label -m metadata -d data`
`fsck`	Check and repair filesystem errors	`-y auto fix -n read only -t type -A all from fstab`

Command	Description	Key Flags / Notes
e2fsck	Check and repair ext2/3/4 filesystems	-y auto fix -n check only -f force -p auto repair
tune2fs	Adjust ext2/3/4 filesystem parameters	-l list -L label -c max mounts -i interval -m reserved
dumpe2fs	Dump ext2/3/4 filesystem information	-h header only
resize2fs	Resize ext2/3/4 filesystem	-p progress -M minimize size
xfs_repair	Repair XFS filesystem	-n no modify -L log reset
xfs_info	Display XFS filesystem geometry information	
xfs_admin	Change parameters of XFS filesystem	-l label -U uuid -L set label
e2label	Set or get ext filesystem label	device newlabel
blkid	Print block device UUID type and label	-s tag -o format -t search

14 RAID and LVM Management

Logical volume and software RAID — enterprise storage

IMPORTANT

Command	Description	Key Flags / Notes
mdadm	Create manage monitor software RAID arrays	--create --assemble --manage --detail --scan --stop -v
pvcreate	Initialize disk as Physical Volume for LVM	-v verbose -f force --dataalignment
pvdisplay	Display Physical Volume information	-v verbose -s short
pvs	Short Physical Volume listing	-o options --units
vgcreate	Create Volume Group from physical volumes	-s PE size -v verbose
vgdisplay	Display Volume Group information	-v verbose -s short
vgs	Short Volume Group listing	-o options
vgextend	Add physical volume to Volume Group	vgname pvpath
lvcreate	Create Logical Volume from Volume Group	-L size -n name -l extents -s snapshot --thin
lvdisplay	Display Logical Volume information	-v verbose -m map
lvs	Short Logical Volume listing	-o options --units
lvextend	Extend a Logical Volume size	-L +size -l +extents -r resize fs
lvreduce	Reduce a Logical Volume size	-L -size -l -extents -r resize fs

Command	Description	Key Flags / Notes
`lvremove`	Remove a Logical Volume	`-f force -v verbose`
`vgremove`	Remove a Volume Group	`-f force`
`pvremove`	Remove Physical Volume label from device	`-f force -v verbose`

15 Firewall Management

Control network access — allow block NAT rules

IMPORTANT

Command	Description	Key Flags / Notes
`iptables`	Set IPv4 firewall rules (netfilter)	`-L list -A append -D delete -I insert -F flush -P policy -t table -j ACCEPT/DROP/REJECT`
`ip6tables`	Set IPv6 firewall rules	`same flags as iptables`
`iptables-save`	Save current iptables rules to stdout	`> rules.txt -t table`
`iptables-restore`	Restore iptables rules from file	`< rules.txt`
`nft`	Modern nftables firewall (replaces iptables)	`list ruleset add rule delete rule flush ruleset`
`ufw`	Uncomplicated Firewall — Ubuntu frontend	`enable disable status allow deny reject limit app list`
`firewall-cmd`	Firewalld frontend — RHEL/CentOS/Fedora	`--permanent --zone --add-service --add-port --reload --list-all`
`ipset`	Manage IP sets for efficient firewall rules	`create add del list destroy test -exist`

16 Log Management and Analysis

Read analyze rotate logs — find errors debug issues

IMPORTANT

Command	Description	Key Flags / Notes
`journalctl`	Query systemd journal logs	`-u unit -f follow -n lines -p priority -b boot --since --until -k kernel -o format`
`tail`	Follow log files in real time	`-f follow -F follow rename -n lines`
`grep`	Search and filter log content	`-i -n -r -l -v --color`
`awk`	Extract specific fields from log lines	`print $1 /pattern/ {action}`
`sed`	Transform and clean log output	`s/old/new/ /pattern/d -n print`

Command	Description	Key Flags / Notes
dmesg	Kernel messages — hardware boot errors	-T timestamps -l err/warn -w follow --clear -H human
logrotate	Rotate compress and manage log files	-v verbose -f force -d debug dry run config file
last	Show login history from wtmp log	-n count -F full time -R no host reboot
lastb	Show bad failed login attempts	-n count -F full time
ausearch	Search Linux audit log records	-m message type -x executable -u user -a event id --start --end
aureport	Summary reports from audit log	-au auth -l login -f file -x exec --start --end

17 Security and Access Control

Harden the system — control access audit restrict

IMPORTANT

Command	Description	Key Flags / Notes
sudo	Run command as superuser or another user	-u user -i login -s shell -l list -k clear cache
visudo	Edit sudoers file safely with syntax check	
chroot	Change root directory for a process	chroot /new/root /bin/bash
ulimit	Set resource limits for user processes	-n files -u procs -m memory -s stack -a all -H hard -S soft
pam	Pluggable Authentication Modules config files	/etc/pam.d/ required sufficient optional sufficient
getfacl	Get Access Control List of a file	-R recursive -p no effective
setfacl	Set Access Control List of a file	-m modify -x remove -R recursive -b remove all u:user:rwx g:group:rx
chattr	Make files immutable or append-only	+i immutable +a append only -R recursive
lsattr	List special file attributes	-R recursive -a all -d dirs
faillock	Manage failed login lock records	--user --reset --list
pwck	Verify integrity of password files	-r read only -s sort -q quiet
grpck	Verify integrity of group files	-r read only -s sort

18 SELinux and AppArmor Management

Mandatory Access Control — enforce security policies

IMPORTANT

Command	Description	Key Flags / Notes
`getenforce`	Show current SELinux enforcement mode	`Enforcing Permissive Disabled`
`setenforce`	Set SELinux to Enforcing or Permissive	`1 enforcing 0 permissive`
`sestatus`	Show detailed SELinux status and policy	`-v verbose -b boolean list`
`semanage`	Manage SELinux policy components	`port fcontext user login boolean -a add -d delete -l list`
`restorecon`	Restore default SELinux file contexts	`-R recursive -v verbose -n dry run -F force`
`chcon`	Change SELinux security context of file	`-t type -r role -u user -R recursive -v verbose`
`getsebool`	Show SELinux boolean values	`-a all booleans`
`setsebool`	Set SELinux boolean values on/off	`-P permanent boolean name on/off`
`audit2allow`	Generate SELinux policy from audit denials	`-a all -m module name -M module -l recent`
`audit2why`	Explain why SELinux denied an action	`-a all -l recent`
`aa-status`	Show AppArmor status and loaded profiles	
`aa-enforce`	Set AppArmor profile to enforce mode	`profile path`
`aa-complain`	Set AppArmor profile to complain mode	`profile path`
`apparmor_parser`	Load AppArmor profiles from file	`-r reload -a add -R remove`

19 System Auditing and Compliance

Track who did what — audit trails for compliance

IMPORTANT

Command	Description	Key Flags / Notes
`auditd`	Linux Audit Daemon — collects audit events	`start stop restart status /etc/audit/auditd.conf`
`auditctl`	Control the Linux audit system	`-l list rules -a add rule -d delete -D delete all -w watch file -p permissions -k key`
`ausearch`	Search audit log for specific events	`-m type -u user -x exe -a event -f file --start --end -i interpret`
`aureport`	Generate reports from audit log data	`--summary -au auth -l login -f file -x exec -m modify --start --end`

Command	Description	Key Flags / Notes
`autrace`	Trace a process using the audit system	`autrace /path/to/program`
`aide`	Advanced Intrusion Detection Environment	`--init --check --update --config`
`last`	Show login audit trail from wtmp	`-n -F -R -x runlevel reboot`
`lastb`	Show bad failed login attempts from btmp	`-n -F -R`
`who`	Show who is currently logged in	`-a all -b boot -H headers -l login processes`
`w`	Show logged-in users with activity info	`-h no header -s short -u cpu usage`
`ac`	Print statistics about user connection times	`-d daily -p per user -y year`

20 Certificate and Encryption

TLS SSL keys hashes — secure data and communications

IMPORTANT

Command	Description	Key Flags / Notes
`openssl`	Swiss army knife for TLS/SSL and encryption	`genrsa req x509 s_client s_server enc dgst verify pkcs12`
`gpg`	GNU Privacy Guard — encrypt sign verify files	`--gen-key --encrypt --decrypt --sign --verify --import --export --list-keys`
`ssh-keygen`	Generate SSH public/private key pairs	`-t ed25519/rsa -b bits -C comment -f file -p passphrase`
`md5sum`	Generate and verify MD5 hash of file	`-c check -b binary *`
`sha256sum`	Generate and verify SHA-256 hash of file	`-c check -b binary *`
`sha512sum`	Generate and verify SHA-512 hash of file	`-c check -b binary *`
`sha1sum`	Generate and verify SHA-1 hash of file	`-c check -b binary`
`base64`	Encode or decode Base64 data	`-d decode -w wrap width -i input -o output`
`xxd`	Create hex dump or reverse hex to binary	`-r reverse -b binary -l length -s skip -c cols`
`certtool`	Generate and manage X.509 certificates	`--generate-privkey --generate-request --generate-certificate`

21 Backup and Recovery

Never lose data — backup sync clone restore systems

IMPORTANT

Command	Description	Key Flags / Notes
rsync	Fast incremental file sync local or remote	-a archive -v verbose -z compress -P progress --delete -n dry run --exclude -e ssh
tar	Create and extract archive files	-c create -x extract -v verbose -f file -z gzip -j bzip2 -J xz -t list -C dir
dd	Low-level byte-for-byte disk clone and backup	if=src of=dst bs=4M status=progress conv=sync,noerror
dump	Backup ext2/3/4 filesystems	-0 full -u update -f file -v verbose level 0-9
restore	Restore from dump backup archives	-i interactive -r restore -t list -f file -v verbose
cpio	Copy files to and from archives	-o create -i extract -p copy -v verbose -d make dirs
cp	Copy with preserve permissions for backup	-a archive -p preserve -r recursive -u update newer
scp	Secure remote copy over SSH	-r recursive -p preserve -P port -C compress
sftp	Interactive secure file transfer	get put mget mput ls cd mkdir rm
rclone	Sync files to remote storage (S3 NFS etc)	copy sync ls check mount --progress --dry-run

22 Network Troubleshooting

Diagnose packet loss latency routing DNS issues

IMPORTANT

Command	Description	Key Flags / Notes
ping	Test ICMP connectivity to a host	-c count -i interval -s packet size -t ttl -W timeout -f flood
traceroute	Trace network path packets take hop by hop	-n no DNS -m max hops -w wait -I ICMP -T TCP -p port
mtr	Combines ping and traceroute — real time	--report --report-cycles -n no DNS -c count --tcp --udp
dig	DNS lookup — query DNS records in detail	@server A AAAA MX NS TXT SOA +short +trace +nocmd -x reverse
nslookup	Query DNS servers interactively or one-shot	server set type=MX set debug
host	Simple DNS lookup command	-t type -a all -v verbose -4 IPv4 -6 IPv6

Command	Description	Key Flags / Notes
nmap	Network port scanner and host discovery	-sV service -sS SYN -p ports -A aggressive -O OS -Pn no ping --script
tcpdump	Capture and analyze network packets	-i interface -n no DNS -w file -r file port host -c count -v verbose
ss	Socket statistics — active connections ports	-t tcp -u udp -l listen -p process -n numeric -a all -s summary
netstat	Network connections stats routing (older)	-t -u -l -p -n -r -s statistics
iperf3	Network bandwidth measurement tool	-s server -c client -t time -b bandwidth -P parallel -R reverse
nc	Netcat — raw TCP/UDP socket connection test	-l listen -p port -v verbose -z scan -u UDP -w timeout
whois	Domain registration and IP block info lookup	whois domain.com whois 8.8.8.8
ethtool	Query network interface hardware settings	-i driver info -s settings -S statistics -k offload

23 Shell Scripting and Automation

Automate repetitive tasks — the engineer superpower

STANDARD

Command	Description	Key Flags / Notes
bash	Run bash shell or execute a bash script	-x debug -e exit on error -u unset error -c command -n syntax check
sh	POSIX-compliant shell for portable scripts	-x trace -e exit error -c command
echo	Print text or variables to terminal	-n no newline -e enable escapes \n \t \a
printf	Formatted print — like C printf	%s string %d int %f float %x hex \n newline
read	Read user input into a variable	-p prompt -s silent -t timeout -n chars -a array
export	Set environment variable for child processes	export VAR=value export -n unset
source	Execute script in current shell context	. file source file
exec	Replace current shell process with a command	exec command exec > file redirect
eval	Execute a string as a shell command	eval "$string"
trap	Handle signals in shell scripts	trap 'cmd' SIGINT trap '' SIGTERM trap - EXIT

Command	Description	Key Flags / Notes
alias	Create shortcut for a command	alias ll='ls -lah' alias -p list unalias
unalias	Remove an alias	-a remove all
set	Set shell options and positional parameters	-x trace -e exit error -u undefined error -o option +o unset
unset	Remove a variable or function from shell	-v variable -f function
declare	Declare variables with attributes	-i integer -r readonly -a array -A assoc -x export -p print
local	Declare local variable inside a function	-i integer -r readonly -a array
readonly	Mark variable as read-only	-p print all -a array -f function
shift	Shift positional parameters to the left	shift n
getopts	Parse command-line options in scripts	getopts ':abc:' opt
test	Evaluate conditional expressions	-f file -d dir -z empty -n not empty -eq -ne -lt -gt
expr	Evaluate arithmetic and string expressions	expr 2 + 2 expr length string expr match expr substr
let	Perform arithmetic in shell scripts	let x=5+3 let x++ let x*=2
seq	Print sequences of numbers	seq 1 10 seq 0 2 20 seq -w seq -s separator
man	Display manual page for any command	-k search -f whatis -a all section number
help	Show help for bash built-in commands	help command
info	GNU info documentation system	info command

24 Cron and Job Scheduling

Schedule tasks to run automatically at any time

STANDARD

Command	Description	Key Flags / Notes
crontab	Manage cron jobs for scheduling tasks	-e edit -l list -r remove -u user crontab -e
at	Schedule a one-time command to run later	at 2pm at now +1 hour at midnight -l list -r remove
atq	List pending at jobs in queue	
atrm	Remove a pending at job by job number	atrm jobnumber
batch	Run job when system load allows	batch < script.sh

Command	Description	Key Flags / Notes
anacron	Run periodic jobs regardless of uptime	-f force -n now -d debug /etc/anacrontab
cron	Daemon that executes scheduled cron jobs	/etc/cron.d/ /etc/cron.daily/ /var/spool/cron/
sleep	Pause execution for a specified time	sleep 5 sleep 2m sleep 1h sleep 0.5
watch	Run command repeatedly at interval	-n seconds -d highlight diff -t no title -e exit on error

25 Environment Variables and Shell Config

STANDARD

Configure shell behaviour and session environment

Command	Description	Key Flags / Notes
env	Show all environment variables or run with env	-i empty env -u unset VAR=val command
printenv	Print specific environment variable value	printenv PATH printenv HOME
export	Set variable available to child processes	export VAR=value export -p list all export -n unset
unset	Remove a variable from the environment	unset VAR -v variable -f function
set	Display or set shell variables and options	-x trace -e exit -u undefined -o pipefail
source	Load variables from a file into current shell	source ~/.bashrc . ~/.profile
bashrc	User shell config file ~/.bashrc	aliases functions PATH PS1 prompt exports
profile	Login shell config ~/.bash_profile ~/.profile	PATH exports run once at login
history	Show command history list	-c clear -d offset -w write -r read HISTSIZE HISTFILE
alias	Create command shortcuts	alias ll='ls -lah' alias -p list all
PS1	Shell prompt variable customization	\u user \h host \w path \$ prompt \t time
PATH	Set executable search path	export PATH=$PATH:/new/dir
ldconfig	Update dynamic linker cache for shared libs	-p print cache -v verbose /etc/ld.so.conf

26 System Boot and Initialization

STANDARD

Control boot process startup targets runlevels

Command	Description	Key Flags / Notes
`systemctl`	Manage systemd targets (replaces runlevels)	`isolate get-default set-default list-units --type=target`
`runlevel`	Show current and previous runlevel	
`init`	Send init signal to change runlevel	`0 halt 1 single 3 multiuser 5 graphical 6 reboot`
`telinit`	Tell init process to change runlevel	`telinit 3 telinit 6`
`dmesg`	Show kernel boot and hardware messages	`-T -l err -w --clear -H`
`journalctl -b`	Show logs from current or past boot	`-b 0 current -b -1 previous --list-boots`
`update-grub`	Update GRUB bootloader config file	`generates /boot/grub/grub.cfg`
`grub-install`	Install GRUB bootloader to disk MBR/EFI	`--target --efi-directory --boot-directory device`
`grub-mkconfig`	Generate GRUB configuration file	`-o /boot/grub/grub.cfg`
`efibootmgr`	Manage UEFI boot entries	`-v verbose -c create -d disk -l loader -b bootnum -B delete -o order`

27 Kernel and Module Management

Load remove inspect kernel modules — hardware drivers

STANDARD

Command	Description	Key Flags / Notes
`uname`	Print system and kernel information	`-r kernel release -a all -s name -v version -m machine -p processor`
`lsmod`	List currently loaded kernel modules	
`modprobe`	Load or remove kernel module with dependencies	`-r remove -v verbose -n dry run -a load multiple -l list`
`rmmod`	Remove a kernel module without dependencies	`-v verbose -f force`
`insmod`	Insert a kernel module directly by file	`insmod module.ko parameters`
`modinfo`	Show information about a kernel module	`modinfo module -p parameters -F field -k kernel`
`depmod`	Generate modules dependency map file	`-a all -v verbose -e report errors kernel version`
`sysctl`	Read and write kernel parameters at runtime	`-a all -w write -p load file net.ipv4 vm.swappiness`
`dkms`	Dynamic Kernel Module Support framework	`add install remove status build autoinstall`

Command	Description	Key Flags / Notes
kexec	Load new kernel to execute without full reboot	-l load -e execute -f force --initrd --command-line

28 Compression and Archiving

STANDARD

Pack unpack compress files for transfer and storage

Command	Description	Key Flags / Notes
tar	Tape Archive — create and extract archives	-c create -x extract -v verbose -f file -z gzip -j bzip2 -J xz -t list -C dir -p preserve
gzip	Compress files using GNU zip algorithm	-d decompress -k keep -v verbose -r recursive -9 max compress -1 fast
gunzip	Decompress gzip compressed files	-k keep -v verbose -r recursive
bzip2	Compress using Burrows-Wheeler algorithm	-d decompress -k keep -v verbose -9 max -1 fast
bunzip2	Decompress bzip2 compressed files	-k keep -v verbose
xz	Compress using LZMA — best ratio	-d decompress -k keep -v verbose -9 max -1 fast -z compress
unxz	Decompress xz files	-k keep -v verbose
zip	Create zip archives — cross-platform	-r recursive -e encrypt -9 max -v verbose -u update -d delete
unzip	Extract zip archives	-l list -d destination -o overwrite -q quiet -v verbose
zcat	View gzip file content without extracting	
zless	Page through gzip file without extracting	
zgrep	Search inside gzip compressed files	same flags as grep
7z	7-Zip — handles many archive formats	a add x extract l list t test e extract flat -p password
cpio	Copy files to and from archive streams	-o create -i extract -p copy -v verbose -d create dirs
ar	Create and manage archive libraries (.a files)	-r replace -x extract -t list -v verbose

29 Swap Memory Management

STANDARD

Manage virtual memory swap space on the system

Command	Description	Key Flags / Notes
swapon	Enable swap space — partition or swap file	-a all from fstab -s show -p priority -v verbose
swapoff	Disable and deactivate swap space	-a all -v verbose
mkswap	Set up swap area on device or file	-L label -U uuid -v version -f force
free	Show total used free swap and RAM	-h human -m MB -g GB -s repeat -t total line
vmstat	Show virtual memory swap and CPU stats	-s summary -d disk interval count
sysctl	Tune swappiness and cache pressure	vm.swappiness=10 vm.vfs_cache_pressure=50

30 Performance Tuning and Optimization

Squeeze maximum performance out of the system

STANDARD

Command	Description	Key Flags / Notes
sysctl	Tune kernel parameters live without reboot	-w vm.swappiness=10 net.core.somaxconn=65535 -p load
ulimit	Set resource limits for user shell processes	-n open files -u processes -m memory -s stack -a all -H hard -S soft
nice	Start process with lower/higher CPU priority	-n value -20 highest priority 19 lowest priority
renice	Change CPU priority of running process	-n value -p pid -u user -g group
ionice	Set disk I/O scheduling class and priority	-c class 0 none 1 real 2 best 3 idle -n priority -p pid
cpupower	Manage CPU frequency scaling governors	frequency-info frequency-set idle-info set
numactl	Control NUMA policy for processes and memory	--hardware --show --membind --cpunodebind --interleave
cgroups	Control groups — limit resources per process	v1 /sys/fs/cgroup v2 unified cpu memory blkio
tuned-adm	Apply pre-built system tuning profiles	list active profile name recommend off
hdparm	Set disk read ahead and DMA settings	-a readahead -d DMA -W write cache -t speed test
blockdev	Call block device ioctls from command line	--setra set readahead --setro read only --getsize64

31 System Troubleshooting and Diagnostics

Debug crashes hangs resource leaks system failures

STANDARD

Command	Description	Key Flags / Notes
`strace`	Trace system calls and signals of a process	`-p PID -e trace=open,read -o file -f forks -t timestamp -c count`
`ltrace`	Trace library calls made by a program	`-p PID -o file -e function -f forks -n indent`
`lsof`	List all open files by processes	`-p PID -i :port -u user -n no DNS +D directory -c command`
`fuser`	Show which process is using a file or port	`-v verbose -k kill -m mount -n network -u user`
`dmesg`	Kernel messages for hardware and driver issues	`--clear -T -l err,crit -w follow -H human --since`
`journalctl`	Systemd journal for service crash logs	`-u service -p err -b boot --since -f follow -n 100`
`sosreport`	Collect system diagnostics into a report	`-- plugin name -o output -a all`
`vmstat`	Show memory CPU I/O stats for diagnosis	`1 10 interval count -s summary -d disk`
`iostat`	Diagnose disk I/O bottlenecks	`-x extended -d disk -h human 1 interval`
`sar`	Historical system stats for trending analysis	`-u cpu -r mem -d disk -n network -q load -b io`
`perf`	Linux profiling — CPU cycles cache misses	`stat record report top annotate list`
`valgrind`	Detect memory leaks and errors in programs	`--tool=memcheck --leak-check=full --track-origins=yes`
`gdb`	GNU Debugger — debug programs and core dumps	`run bt backtrace p print b breakpoint c continue q quit`
`core dump`	Configure and analyse core dump files	`ulimit -c unlimited /proc/sys/kernel/core_pattern`

32 Network File Sharing

Share files across network — NFS Samba CIFS

STANDARD

Command	Description	Key Flags / Notes
`exportfs`	Export NFS directories to clients	`-a all -r re-export -u unexport -v verbose /etc/exports`

Command	Description	Key Flags / Notes
showmount	Show NFS server mount info and exports	-e exports -a all mounts -d directories host
mount.nfs	Mount NFS network filesystem	-o options rw ro vers=4 noatime soft hard
mount.cifs	Mount Samba/Windows CIFS share	-o user= password= domain= uid= gid= vers=
smbclient	Access Samba/Windows shares from CLI	-L list shares -U user //host/share get put ls
nmblookup	NetBIOS name query for Samba hosts	-A IP -U broadcast -R use WINS name
nfsstat	Show NFS statistics and activity	-s server -c client -m mount -z reset
rpcinfo	Report RPC information for NFS services	-p host -s short -T transport

33 DNS and Hostname Management

Resolve query and manage DNS names and records

STANDARD

Command	Description	Key Flags / Notes
dig	Detailed DNS record lookup tool	@server A AAAA MX NS TXT SOA +short +trace +noall +answer -x reverse -4 -6
nslookup	Query DNS servers interactively	server 8.8.8.8 set type=MX set debug name
host	Simple DNS lookup command	-t type -a all -v verbose -l zone transfer -4 -6
hostname	Show or set the system hostname	-I all IPs -f FQDN (Fully Qualified Domain Name) -s short -d domain
hostnamectl	Set hostname with systemd (persistent)	set-hostname name status
resolvectl	Resolve DNS queries via systemd-resolved	query status statistics flush-caches dns
nsswitch.conf	Name service switch config file	/etc/nsswitch.conf hosts: files dns order matters
resolv.conf	DNS resolver configuration file	/etc/resolv.conf nameserver search domain options
hosts	Static hostname to IP mapping file	/etc/hosts 127.0.0.1 localhost IP hostname
named	BIND DNS server daemon	start stop status /etc/named.conf /var/named/
rndc	Remote Name Daemon Control for BIND	reload flush status querylog stats

34 Time and Date Management

Sync system time — critical for logs certificates cron

STANDARD

Command	Description	Key Flags / Notes
`date`	Display or set system date and time	`+%Y-%m-%d +%H:%M:%S -s set -u UTC -d date string`
`timedatectl`	Query and change system time via systemd	`status set-time set-timezone set-ntp list-timezones`
`hwclock`	Access hardware real-time clock (RTC)	`--show --set --systohc --hctosys --utc --localtime`
`ntpdate`	Set system time from NTP server (one-shot)	`-u unburst -q query -v verbose server`
`chronyc`	Control chronyd NTP daemon interactively	`tracking sources sourcestats activity makestep offline`
`ntpq`	Monitor NTP daemon status and peers	`-p peers -i interactive -n numeric -c command`
`cal`	Display calendar in terminal	`-m monday first -3 three months -y full year -j julian`
`tzselect`	Interactively select a timezone	
`zdump`	Print current time in specified timezone	`zdump America/New_York -v verbose`

35 System Shutdown and Power Management

Safely shutdown reboot suspend power off server

STANDARD

Command	Description	Key Flags / Notes
`shutdown`	Schedule system shutdown or reboot	`-h halt -r reboot -P poweroff now +minutes time message -c cancel`
`reboot`	Reboot the system immediately	`-f force -p poweroff --halt`
`halt`	Stop the CPU without powering off	`-f force -p poweroff -d no sync`
`poweroff`	Power off the system completely	`-f force --halt --no-wall`
`systemctl`	Suspend hibernate reboot via systemd	`suspend hibernate hybrid-sleep reboot poweroff halt`
`wall`	Broadcast message to all logged-in users	`wall 'system going down in 5 minutes'`
`notify-send`	Send desktop notification (GUI systems)	`-t timeout -u urgency -i icon title message`
`pm-suspend`	Suspend system to RAM (older method)	

Command	Description	Key Flags / Notes
`powertop`	Analyze power consumption and wake-ups	`--auto-tune --html=report.html --calibrate`

36 Hardware Information and Device Management

STANDARD

Identify hardware — CPU RAM PCI USB drives

Command	Description	Key Flags / Notes
`lshw`	Detailed hardware information report	`-short -class cpu/memory/network -html -json -businfo`
`lspci`	List all PCI devices and their drivers	`-v verbose -k kernel driver -nn IDs -t tree -s slot`
`lsusb`	List all USB devices connected	`-v verbose -t tree -s bus:dev -d vendor:product`
`dmidecode`	Read hardware info from BIOS DMI/SMBIOS	`-t type 0 bios 1 system 4 processor 17 memory -s string`
`lscpu`	Detailed CPU architecture information	`-e extended -p parse -J json -x hex`
`lsmem`	List memory ranges and their properties	`-a all -o output --json --summary`
`lsblk`	List block storage devices and partitions	`-f filesystem -o NAME,SIZE,TYPE,UUID -t tree -J json`
`hwinfo`	Detailed hardware probe information	`--cpu --memory --disk --network --all --short`
`inxi`	System information script — human readable	`-F full -C cpu -M machine -N network -D disk -G graphics`
`udevadm`	Query and manage udev device events	`info monitor trigger settle control test`
`dmesg`	Hardware detection messages from kernel	`-T timestamps -l info grep device --clear`
`sensors`	Show hardware temperature and voltage	`sensors-detect first -f fahrenheit -A no labels`
`nvidia-smi`	NVIDIA GPU information and monitoring	`-q query -l loop --format=csv --query-gpu=`

37 Terminal Multiplexing and Sessions

STANDARD

Persistent sessions — survive disconnects run parallel

Command	Description	Key Flags / Notes
`tmux`	Terminal multiplexer — panes windows sessions	`new attach ls kill-session -s name Ctrl+b prefix key`
`screen`	Screen multiplexer — detach reattach sessions	`-S name -r reattach -ls list -d detach -X command`
`byobu`	Friendly wrapper around tmux or screen	`F2 new window F3 prev F4 next F6 detach F9 config`
`nohup`	Keep process running after terminal logout	`nohup command & output to nohup.out`
`disown`	Remove job from shell job table	`disown %1 disown -h disown -a`

38 User Session and Login Management

Track who is on system — manage active sessions

STANDARD

Command	Description	Key Flags / Notes
`who`	Show currently logged-in users	`-a all -b boot time -H headers -l login -q count`
`w`	Show users and what they are currently doing	`-h no header -s short -u cpu -i no idle`
`last`	Show login history from wtmp log file	`-n count -F full time -R no hostname -x runlevel reboot`
`lastlog`	Show most recent login of every user	`-u user -t days -b before days`
`lastb`	Show bad failed login attempts btmp	`-n count -F full time -R no host`
`wall`	Send message broadcast to all terminals	`wall 'message text' echo msg | wall`
`write`	Send message to specific user terminal	`write username [tty]`
`mesg`	Allow or deny messages from other users	`mesg y mesg n mesg status`
`loginctl`	Manage systemd login sessions and users	`list-sessions list-users show-session terminate-session lock-session`
`pkill`	Kill all processes of a user	`-u username -9 force -t terminal`

39 Multi-user Environment and Quota Management

Disk quotas resource limits for shared server users

STANDARD

Command	Description	Key Flags / Notes
quota	Show disk quota usage for a user	-u user -g group -v verbose -s human -l local
quotaon	Enable disk quotas on a filesystem	-a all -u user -g group -v verbose
quotaoff	Disable disk quotas on a filesystem	-a all -u user -g group -v verbose
edquota	Edit quota limits for a user or group	-u user -g group -p prototype -t grace time
repquota	Summary of disk quotas for a filesystem	-a all -u user -g group -v verbose -s human
quotacheck	Scan filesystem to build quota files	-a all -u user -g group -v verbose -m no remount
setquota	Set quota limits from command line	user filesystem bsoftt bhard isoftt ihard
ulimit	Set per-shell resource limits	-n files -u procs -m memory -v virtual -a all -H hard -S soft

40 Rescue and Recovery Operations

Fix broken systems — unbootable filesystems corrupted data

STANDARD

Command	Description	Key Flags / Notes
fsck	Check and repair filesystem	-y auto fix -n read only -A all fstab -t type -v verbose
e2fsck	Repair ext2/3/4 filesystems	-y -n -f force -p auto -c bad blocks
testdisk	Recover lost partitions and fix boot sectors	interactive Analyse Write Boot Undelete
photorec	Recover deleted files by file signature	interactive select filesystem file types
chroot	Chroot into broken system for repair	mount --bind /dev /mnt/dev chroot /mnt /bin/bash
grub-rescue	Repair GRUB from rescue prompt	set prefix set root insmod normal normal
init 1	Boot into single-user rescue mode	telinit 1 systemctl rescue
dd	Clone disk for forensic or recovery copy	if=/dev/sda of=backup.img bs=4M status=progress conv=noerror
mount -o ro	Mount filesystem read-only for examination	mount -o ro,noload /dev/sda1 /mnt
lsblk	Identify all available block devices	-f show filesystem -o UUID,NAME,SIZE,TYPE

41 File Editing — Terminal Editors

Edit config files scripts logs directly in terminal

STANDARD

Command	Description	Key Flags / Notes
vim	Vi Improved — powerful modal text editor	i insert Esc normal :w save :q quit :wq dd delete yy copy p paste /search :%s/old/new/g
vi	Original Vi editor — present on every Unix	i insert Esc :w :q :wq dd yy /search
nano	Simple beginner-friendly terminal editor	Ctrl+O save Ctrl+X exit Ctrl+W search Ctrl+K cut Ctrl+U paste
ed	Original line editor — POSIX standard	p print d delete s/old/new/ w write q quit
emacs	Full-featured editor and environment	Ctrl+x Ctrl+s save Ctrl+x Ctrl+c quit Ctrl+s search Alt+x command
tee	Write to file and stdout at same time	command | tee file.txt -a append
sed -i	In-place file editing using sed	sed -i 's/old/new/g' file -i.bak backup first

42 Performance Benchmarking and Testing

Measure disk CPU network RAM throughput and latency

STANDARD

Command	Description	Key Flags / Notes
dd	Measure raw disk read write throughput	if=/dev/zero of=test bs=1G count=1 oflag=direct status=progress
hdparm	Measure disk read speed and parameters	-t timing read -T cache read --direct -v verbose
fio	Flexible I/O tester — disk benchmarking	--name --rw=read/write/randread --bs --size --numjobs --iodepth
iperf3	Network bandwidth measurement between hosts	-s server -c client -t duration -b bandwidth -P streams -R reverse
stress	Impose load on CPU memory disk I/O	--cpu N --vm N --vm-bytes --io N --timeout
sysbench	System benchmark for CPU memory file mutex	cpu memory fileio threads run --threads --time
time	Measure real user and system time of a command	time command real=wall clock user=CPU time sys=kernel
ab	Apache Bench — HTTP server load testing	-n requests -c concurrent -t timelimit -H header -p POST file

Command	Description	Key Flags / Notes
vmstat	CPU and memory benchmark baseline view	1 5 interval count -d disk stats

43 Incident Response and Forensics

Investigate breaches analyze what happened on system

STANDARD

Command	Description	Key Flags / Notes
last	Show login history — detect unauthorized access	last last reboot last username -F full time -n 50
lastb	Show failed login attempts from btmp log	-n count -F -R -a hostname username
ausearch	Search audit logs for suspicious events	-m LOGIN -m SYSCALL -u user -x executable --start today
aureport	Summarize audit events for review	-au auth -l login -f file access -x exec --failed
lsof	Find what files/sockets a process has open	-p PID -i :port -u user -n -P +D /var
ss	List active network connections and sockets	-t -u -l -p -n -a 'sport = :22'
tcpdump	Capture network traffic for analysis	-i eth0 -w capture.pcap -n port 80 host IP -c count
netstat	Check for unexpected listening ports	-tlnp listening TCP with PID -ulnp UDP
find	Find recently modified files for investigation	-mtime -1 -newer reference -type f -perm -4000 SUID
ps	Spot unexpected processes running	-aux -ef --forest grep suspicious
who	See who is currently logged in right now	-a -H last w
dmesg	Check kernel logs for rootkit signs	-T -l err,crit grep unusual --since
md5sum	Verify file integrity against known hash	md5sum file md5sum -c checksums.txt
sha256sum	Verify cryptographic hash of binary files	sha256sum file -c check --quiet

44 Printing and Output Management

Manage print jobs and printer queues from terminal

STANDARD

Command	Description	Key Flags / Notes
`lp`	Print a file to default or named printer	`-d printer -n copies -P pages -o options`
`lpr`	Print file using BSD printing interface	`-P printer -# copies -o option`
`lpq`	Show print queue status	`-P printer -a all printers -l long`
`lprm`	Remove a print job from queue	`-P printer job-id - all jobs`
`lpstat`	Show printer and job status	`-p printers -a accepting -d default -s summary -t all`
`cancel`	Cancel print jobs	`cancel job-id -a all -u user`
`cupsctl`	Configure CUPS printing system	`--debug-logging --no-debug-logging --share-printers`

45 Storage Networking — NFS iSCSI

Connect to SAN and networked storage infrastructure

STANDARD

Command	Description	Key Flags / Notes
`iscsiadm`	Manage iSCSI initiator — connect to SAN	`-m discovery -m node --login --logout -T target -p portal --op`
`targetcli`	Configure iSCSI target (server side)	`ls /backstores /iscsi create set delete saveconfig`
`multipath`	Manage multipath devices for redundancy	`-l list -ll verbose -f flush -r reload /etc/multipath.conf`
`multipathd`	Multipath daemon — monitors path state	`show paths show maps reconfigure resize map`
`exportfs`	Export NFS directories to network clients	`-a all -r re-export -u unexport -v verbose`
`nfsstat`	Show NFS statistics client and server	`-s server -c client -m mounts -z reset -v verbose`
`rpcinfo`	Show RPC program info for NFS portmapper	`-p host -s short -T protocol`

46 Package Management — Universal Layer

Install update remove software at system level

STANDARD

Command	Description	Key Flags / Notes
`dpkg`	Debian/Ubuntu low-level package management	`-i install -r remove -P purge -l list -L files -s status --configure -a`

Command	Description	Key Flags / Notes
`rpm`	Red Hat/CentOS low-level package management	`-i install -e erase -q query -l list files -a all -v --nodeps --force`
`dpkg-query`	Query installed dpkg packages	`-W format -l list -s status -L listfiles -p info`
`snap`	Install and manage snap packages	`install remove list info refresh find run connect`
`flatpak`	Install universal sandboxed applications	`install uninstall list info update run remote-add`
`pip`	Python package installer	`install uninstall list freeze show search install -r`
`pip3`	Python 3 package installer	`install uninstall list freeze -r requirements.txt`
`gem`	Ruby package manager for gems	`install uninstall list search update cleanup`
`npm`	Node.js package manager	`install uninstall list update init run global -g`
`cargo`	Rust package manager	`install uninstall build run test update search`

47 Disk Health and SMART Monitoring

Predict drive failures before data loss occurs

STANDARD

Command	Description	Key Flags / Notes
`smartctl`	Query disk SMART health data	`-i info -H health -a all -t short -t long -l log -x all extended`
`hdparm`	Test disk speed and view parameters	`-t timing -T cache -I info -d DMA -W write cache`
`badblocks`	Scan disk for bad sectors	`-v verbose -w destructive write -n non-destructive -b block size -o output`
`fsck`	Check filesystem and fix bad block references	`-c check bad blocks -y auto fix -t type -A all`
`lsblk`	List drives with model and size info	`-o NAME,SIZE,MODEL,SERIAL,TYPE,TRAN`
`blkid`	Show filesystem UUID and type of drives	`-o list -s UUID -s TYPE`

48 Miscellaneous System Utilities

Essential utilities that do not fit one category

STANDARD

Command	Description	Key Flags / Notes
`cal`	Display calendar for month or year	`-m monday -3 three months -y year -j julian -w week num`
`bc`	Command-line calculator — arbitrary precision	`-l math library scale=2 ibase obase sqrt s c`
`factor`	Print prime factors of a number	`factor 360`
`xargs`	Build commands from standard input pipeline	`-n max -I replace -P parallel -0 null -r no run if empty`
`yes`	Output y continuously — automate confirmations	`yes yes no yes string`
`true`	Return exit code 0 — always succeeds	`used in scripts and loops`
`false`	Return exit code 1 — always fails	`used in scripts for testing`
`nohup`	Detach process from terminal session	`nohup command & output to nohup.out`
`tput`	Query terminal capabilities and set colors	`tput cols tput lines tput setaf 1 tput bold tput sgr0`
`stty`	Change and print terminal line settings	`-a all sane reset rows cols speed echo -echo`
`clear`	Clear the terminal screen	`Ctrl+L shortcut`
`reset`	Reset terminal to sane state	`useful when terminal gets corrupted`
`script`	Record terminal session to a file	`-a append -q quiet -t timing scriptreplay`
`scriptreplay`	Replay a recorded terminal session	`scriptreplay -t timing session`
`dd`	General purpose data conversion and copy	`if= of= bs= count= skip= seek= conv=`
`printf`	Format and print data to terminal	`%s %d %f %x %o \n \t -v var`
`uuidgen`	Generate a universally unique identifier UUID	`-r random -t time based`
`tee`	Split output to file and stdout simultaneously	`-a append -i ignore interrupt`

49 Kernel Debugging and Advanced Diagnostics

Deep system internals — kernel panics crashes profiling

STANDARD

Command	Description	Key Flags / Notes
perf	Linux performance profiling tool	stat record report top annotate list probe trace
gdb	GNU Debugger for programs and core dumps	run bt p b c n s watch info quit core file
crash	Analyze Linux kernel crash dumps	crash vmlinux vmcore bt ps sym dis log
kdump	Capture kernel crash dump on panic	/etc/kdump.conf crashkernel= in grub systemctl kdump
kgdb	Kernel GNU Debugger — live kernel debug	requires serial or KDB CONFIG_KGDB=y
sysrq	Magic SysRq key — emergency kernel control	echo b > /proc/sysrq-trigger reboot s sync u remount
dmesg	Kernel ring buffer — hardware driver messages	--clear -T -l debug,info,warn,err,crit
proc	Virtual filesystem exposing kernel internals	/proc/cpuinfo meminfo net sys PID/ version mounts
sys	Kernel sysfs virtual filesystem	/sys/class/ /sys/block/ /sys/bus/ /sys/devices/
kallsyms	All kernel symbol addresses	cat /proc/kallsyms grep symbol name

50 Remote File Transfer

Transfer files securely between local and remote systems

STANDARD

Command	Description	Key Flags / Notes
scp	Secure Copy over SSH	-r recursive -P port -p preserve -C compress -i key -v verbose
sftp	Secure FTP — interactive SSH file transfer	-P port -i key get put mget mput ls cd mkdir rm
rsync	Efficient incremental sync local or remote	-a -v -z -P --delete --exclude -e ssh --bwlimit -n dryrun
wget	Download files from HTTP HTTPS FTP servers	-O output -r recursive -P prefix -c continue -q quiet --limit-rate
curl	Transfer data with URL support multiprotocol	-o -O -L -I -X -d -H -u -k --upload-file -T -F
ftp	File Transfer Protocol client	open get put mget mput ls cd binary ascii bye
nc	Netcat raw file transfer over TCP	nc -l port > file nc host port < file

COMPLETE SUMMARY

Priority	Categories	Commands	Use Case
CRITICAL	5	67	Use every single day — core of all Linux work
HIGH	5	86	Use almost every day — server and admin work
IMPORTANT	12	145	Regular server management and operations
STANDARD	28	292	Periodic tasks — advanced and specialized
TOTAL	50	590	**Complete universal Linux command reference**

MASTERY ROADMAP

Level 1 — Beginner (Get your first Linux job)

Categories 1-5: File management, Text processing, Searching, Permissions, File viewing

Level 2 — Junior Engineer (Survive daily work)

+ Categories 6-10: Process management, Users, Monitoring, Networking, SSH

Level 3 — Mid-level / Sysadmin (Manage production servers)

+ Categories 11-22: Services, Storage, Filesystem, Firewall, Security, Logs, Backup

Level 4 — Senior / DevOps Engineer (Full system control)

+ Categories 23-50: Scripting, Scheduling, Kernel, Forensics, Benchmarking, Recovery